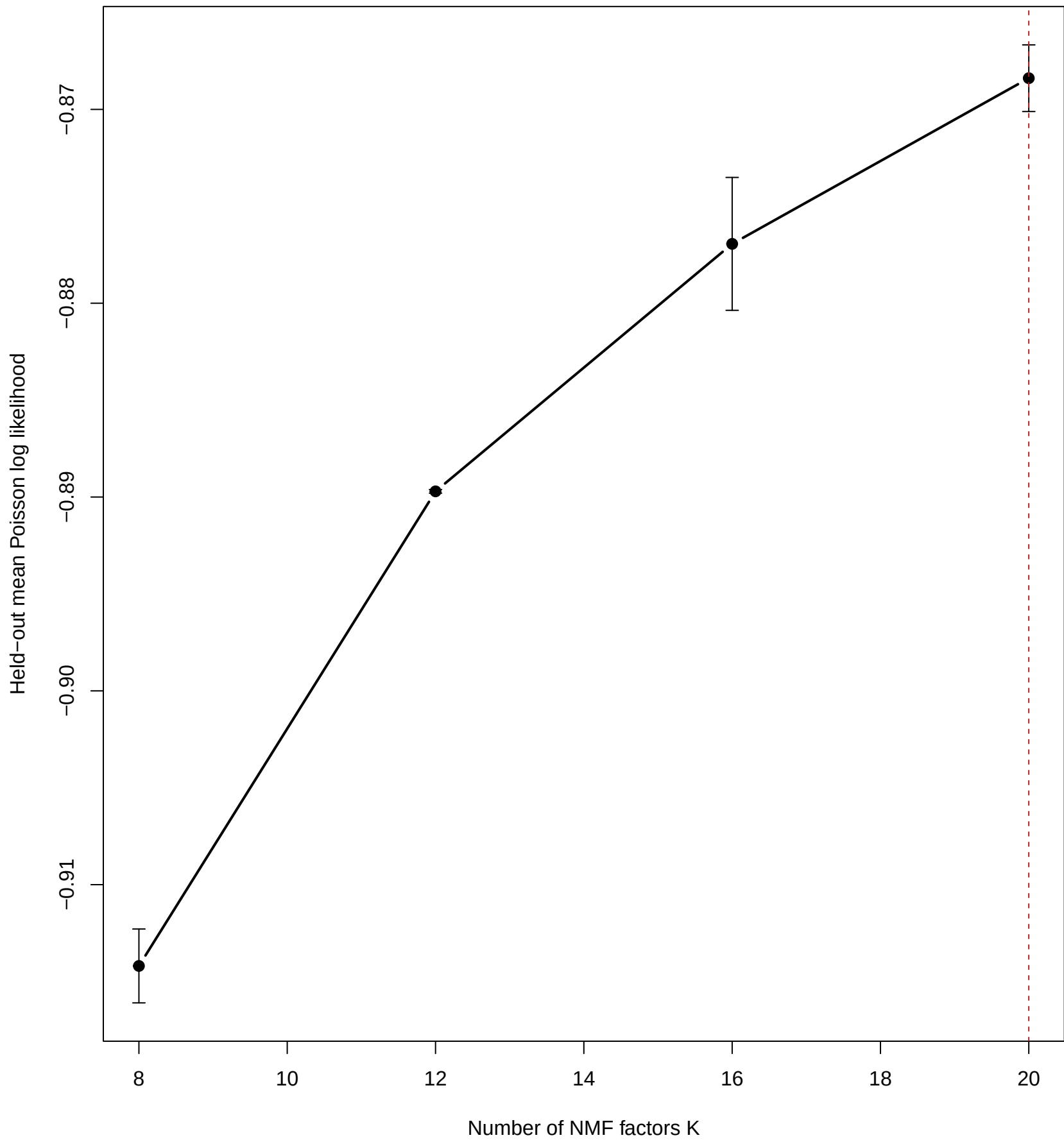
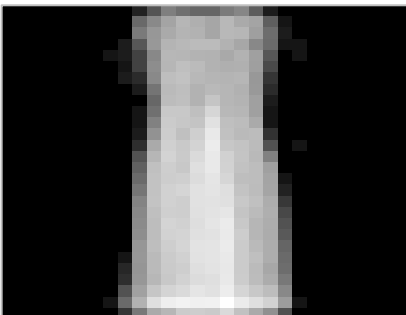


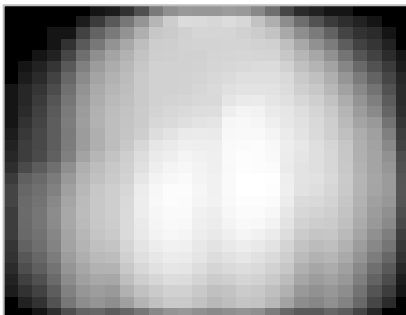
Held-out selection chooses K = 20



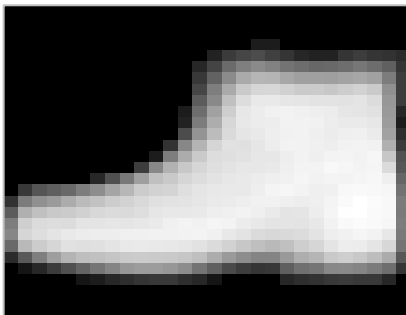
F1: Dress



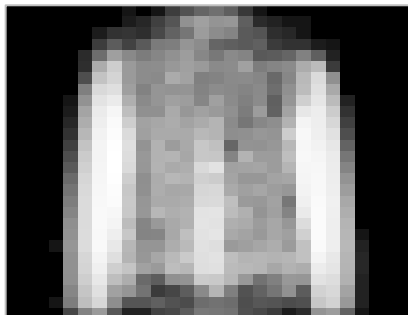
F2: T-shirt/top



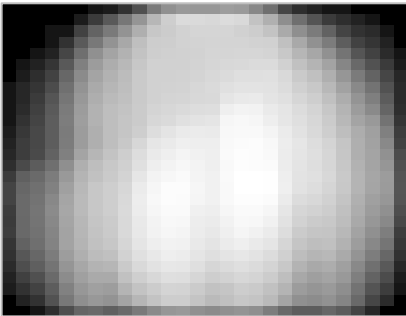
F3: Ankle boot



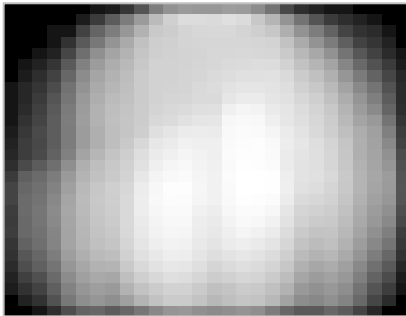
F4: Coat



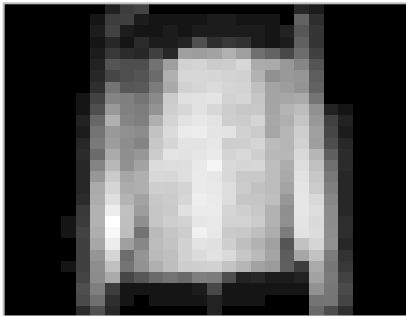
F5: T-shirt/top



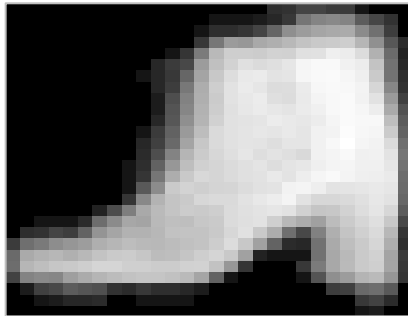
F6: T-shirt/top



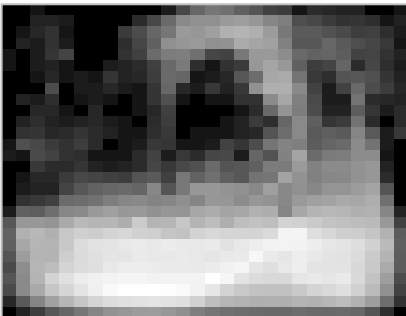
F7: Dress



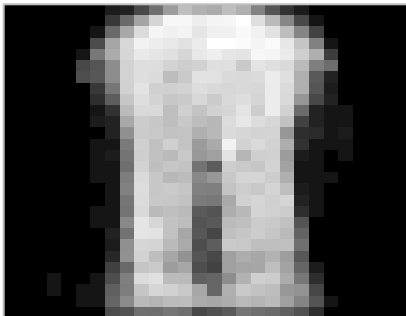
F8: Ankle boot



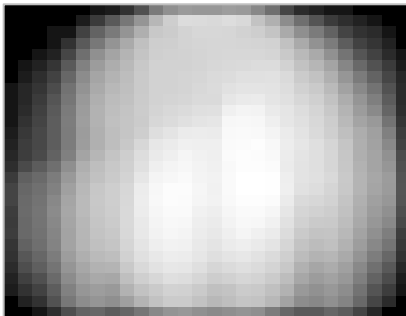
F9: Bag



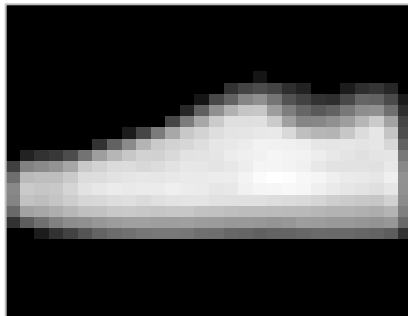
F10: Coat



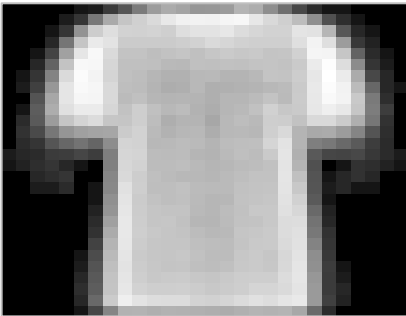
F11: T-shirt/top



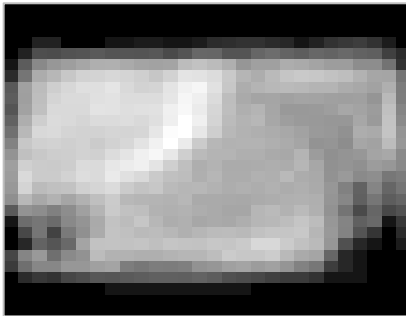
F12: Sneaker



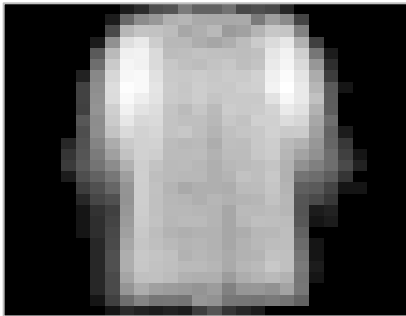
F13: T-shirt/top



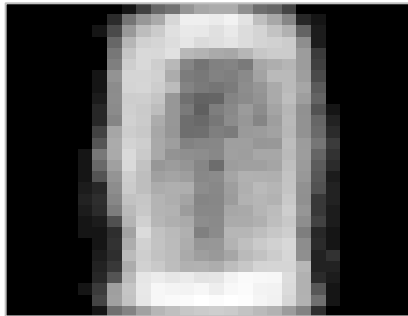
F14: Bag



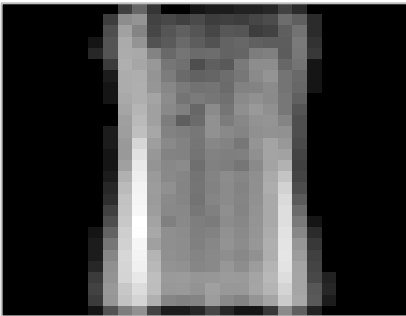
F15: T-shirt/top



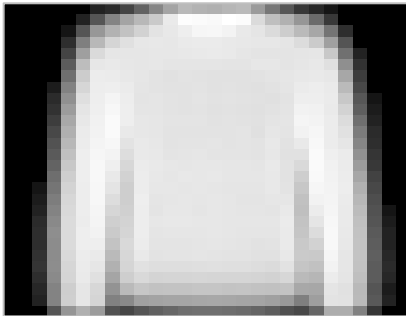
F16: Dress



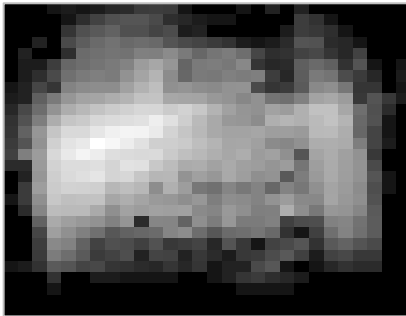
F17: Dress



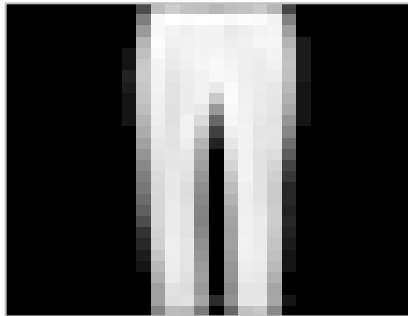
F18: Pullover



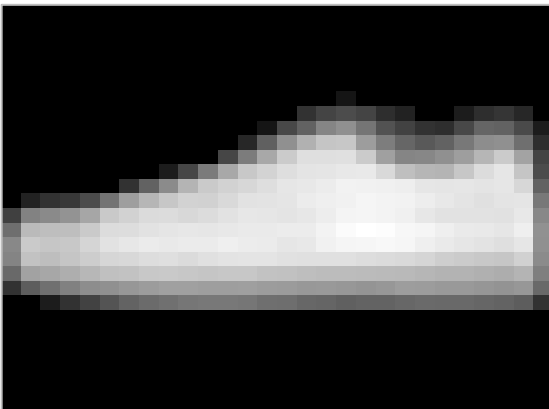
F19: Bag



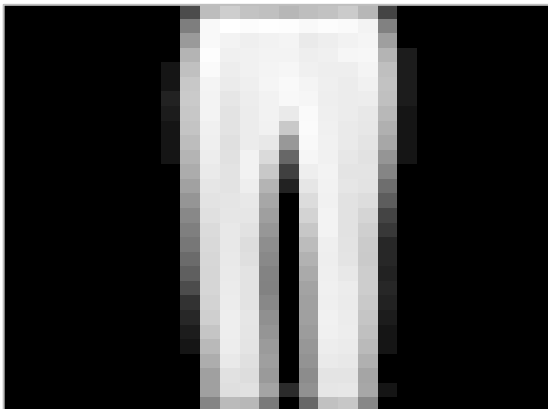
F20: Trouser



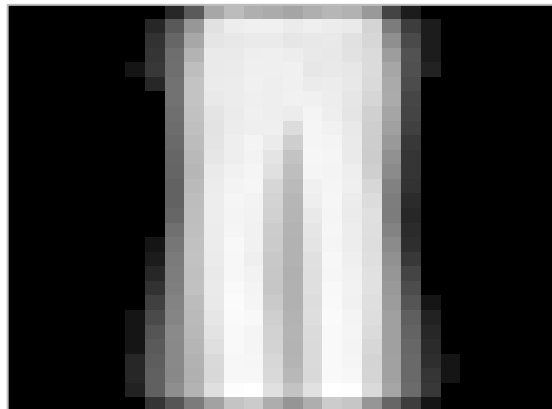
S1 (D90=1): Sneaker



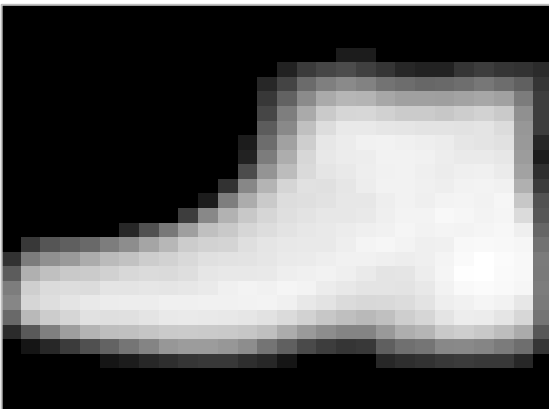
S2 (D90=1): Trouser



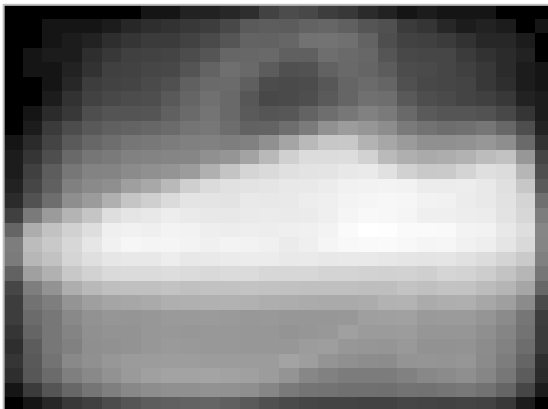
S3 (D90=3): Trouser



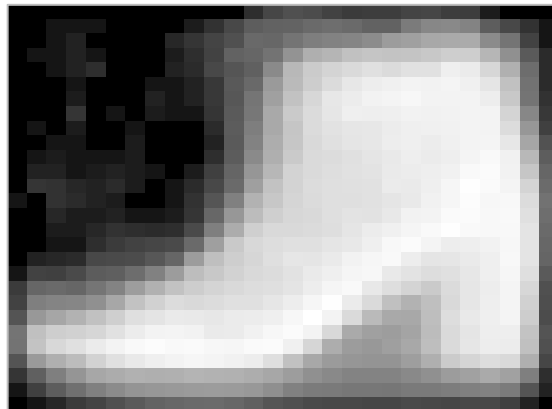
S4 (D90=1): Ankle boot



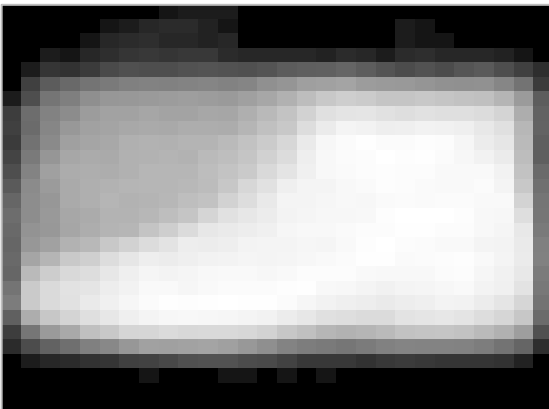
S5 (D90=3): Sneaker



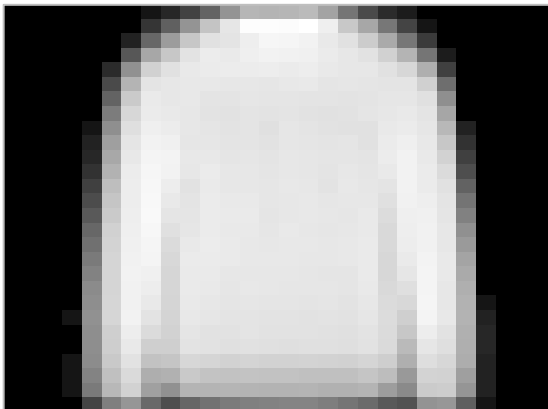
S6 (D90=2): Ankle boot



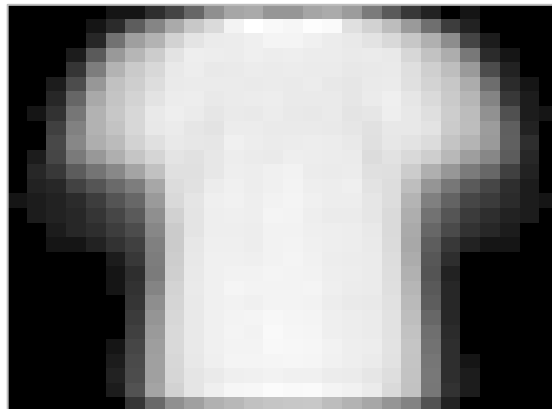
S7 (D90=2): Bag



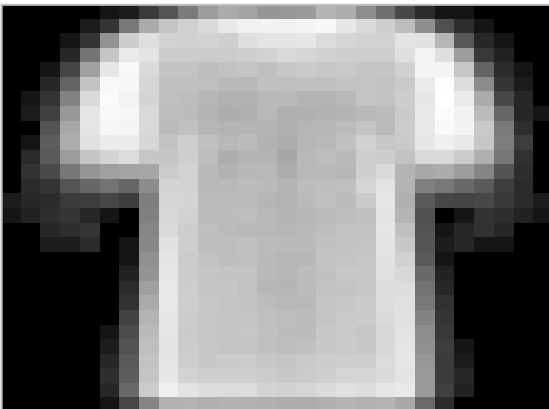
S8 (D90=2): Coat



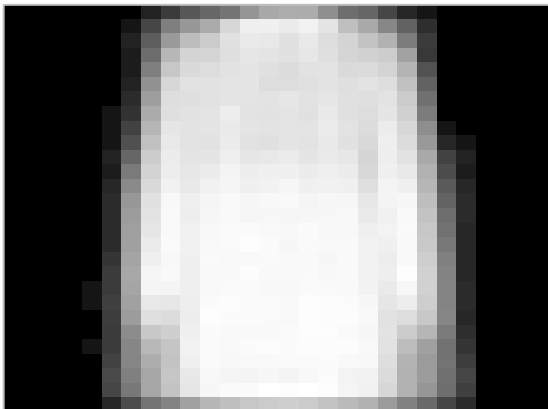
S9 (D90=3): T-shirt/top



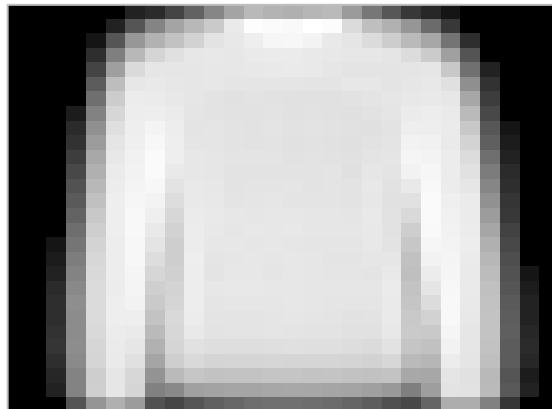
S10 (D90=1): T-shirt/top



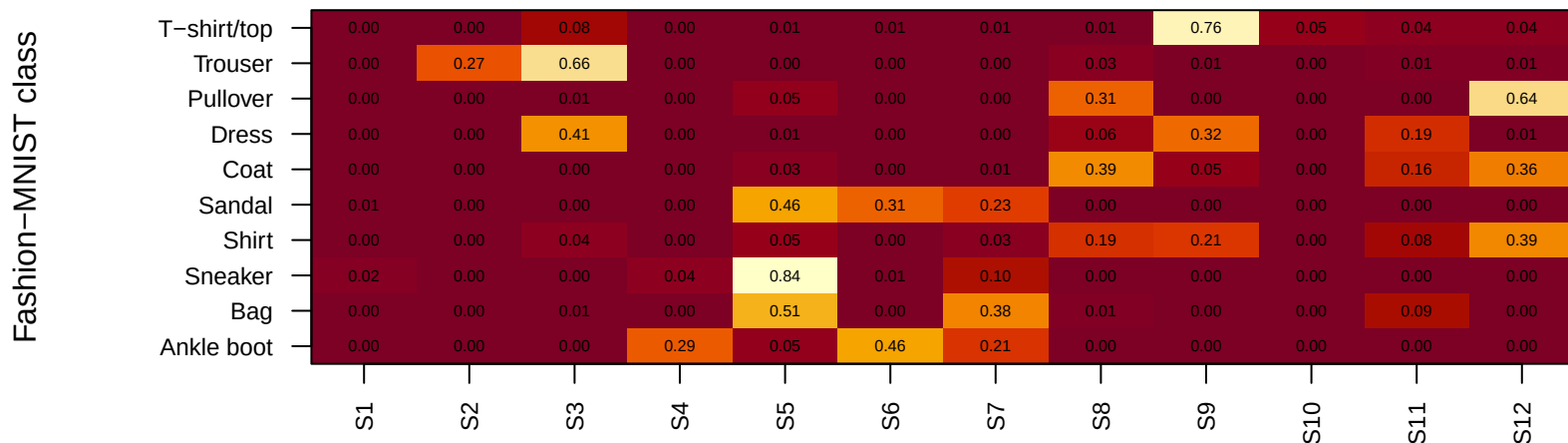
S11 (D90=2): Dress



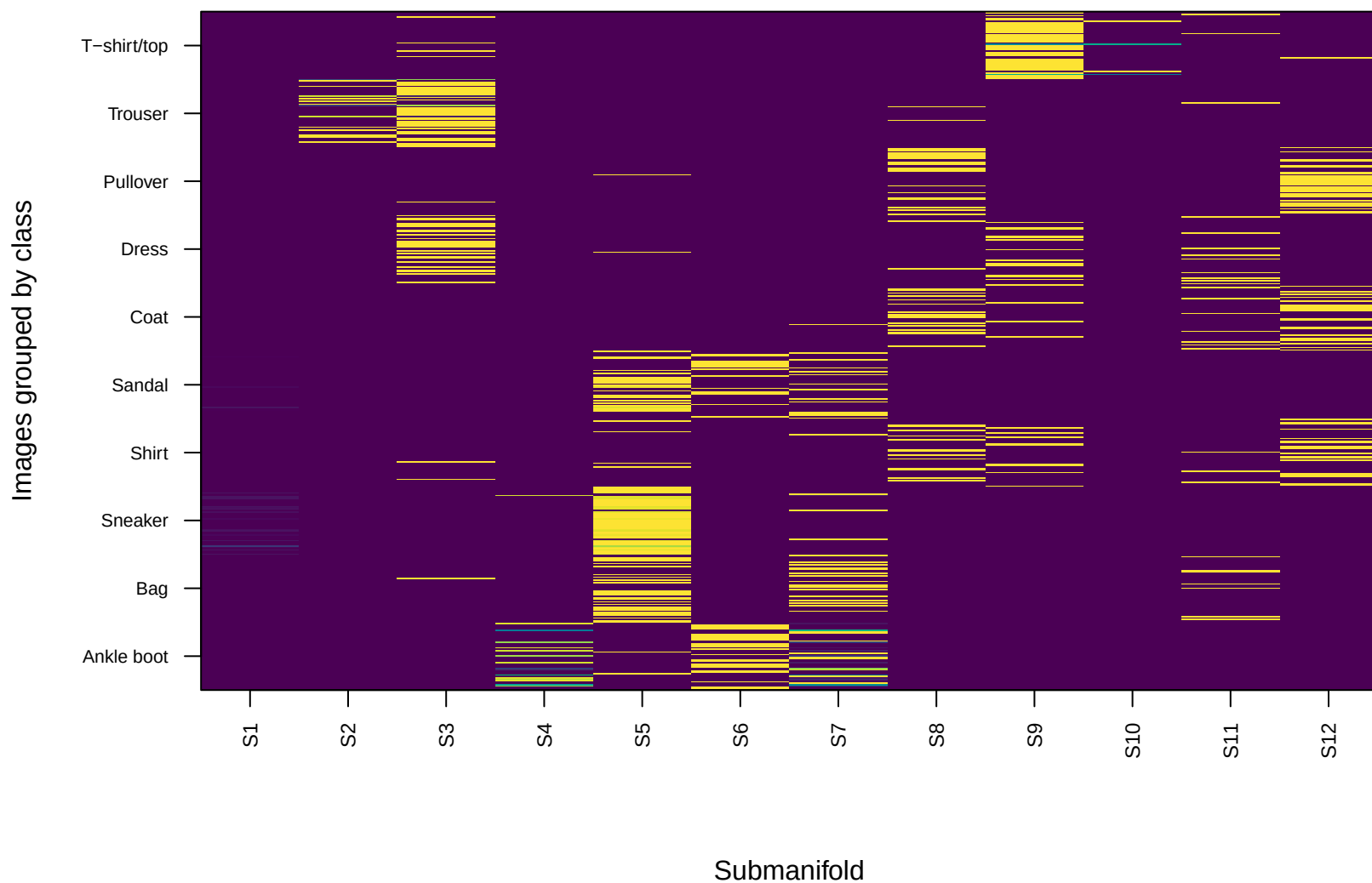
S12 (D90=1): Pullover



Average submanifold responsibility within each clothing class



Individual responsibilities (up to 50 images per class)

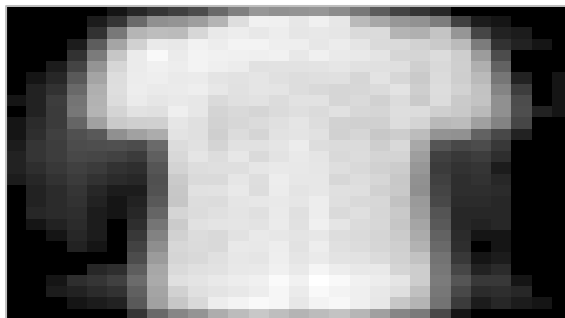


Representative images and reconstructions

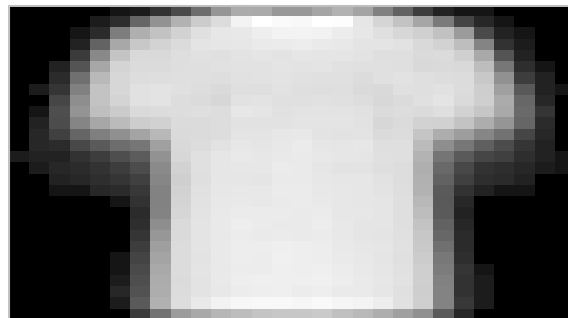
T-shirt/top: observed



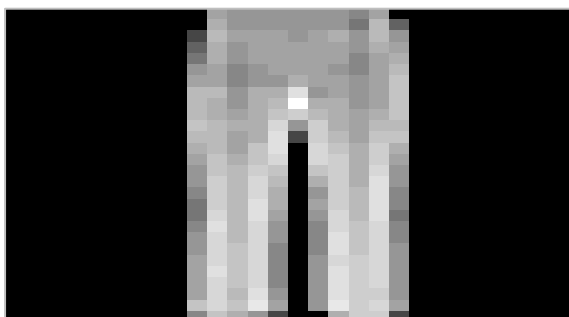
Vanilla NMF



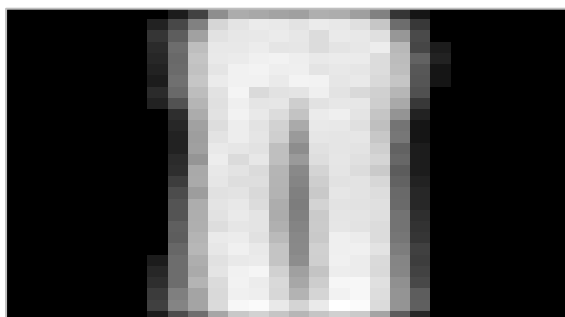
MiSo



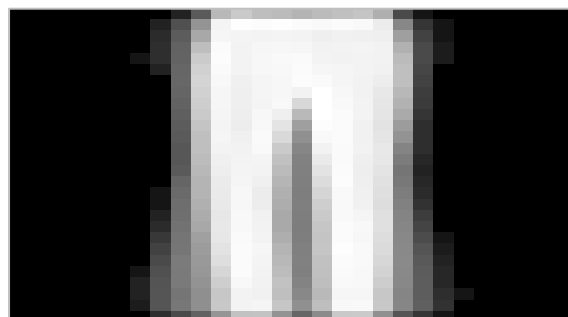
Trouser: observed



Vanilla NMF



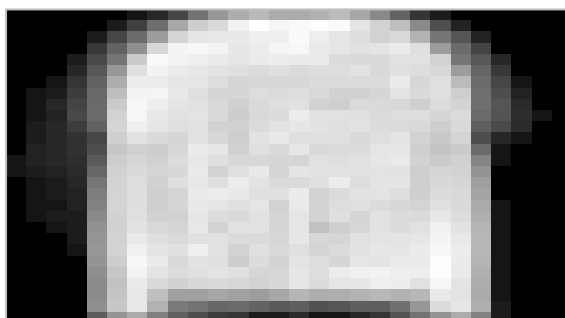
MiSo



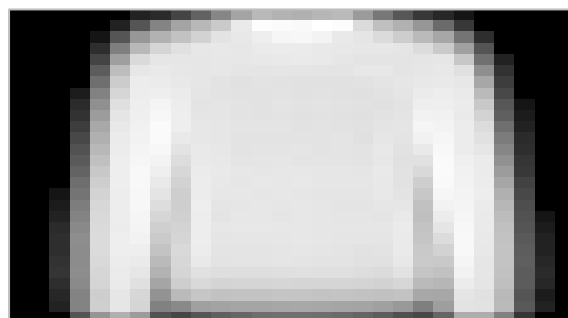
Pullover: observed



Vanilla NMF



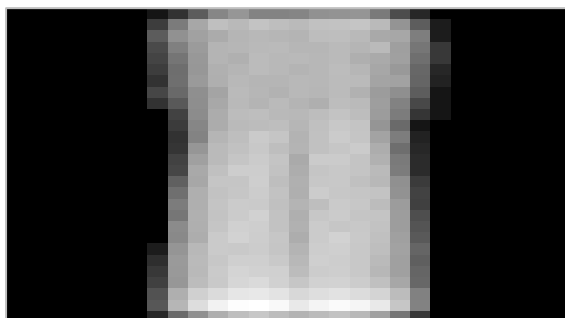
MiSo



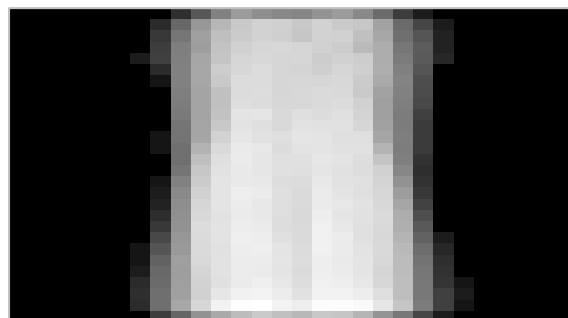
Dress: observed



Vanilla NMF



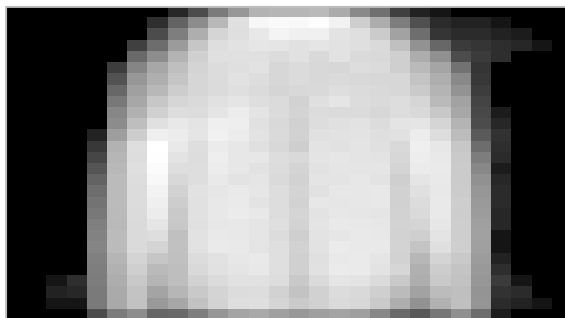
MiSo



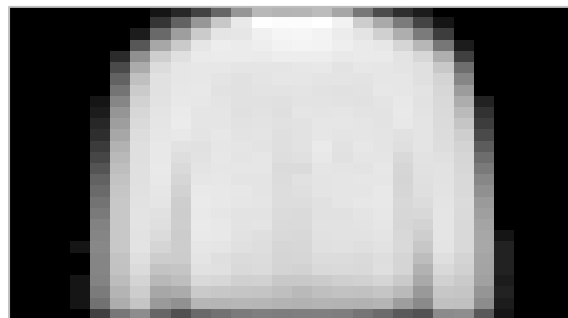
Coat: observed



Vanilla NMF

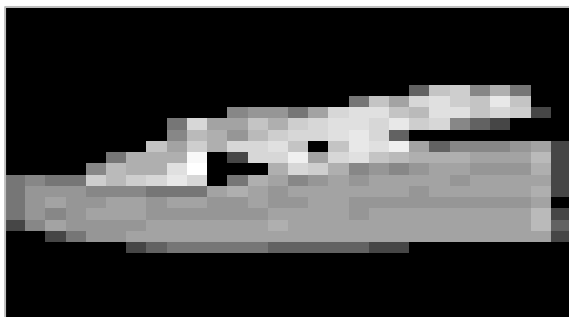


MiSo

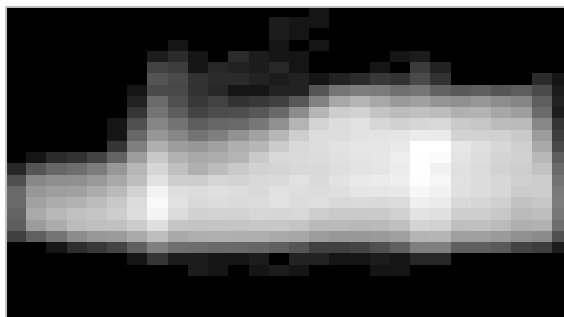


Representative images and reconstructions

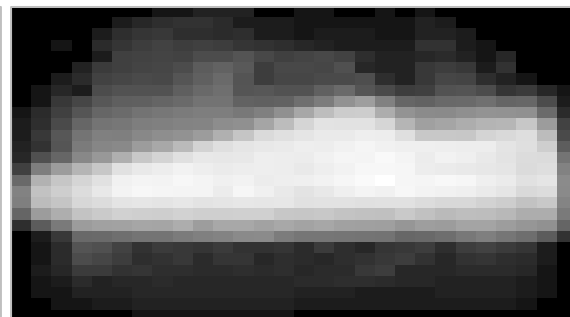
Sandal: observed



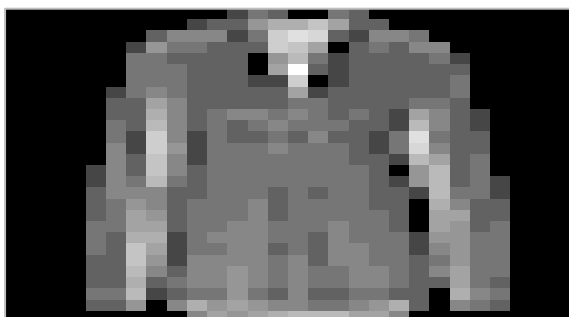
Vanilla NMF



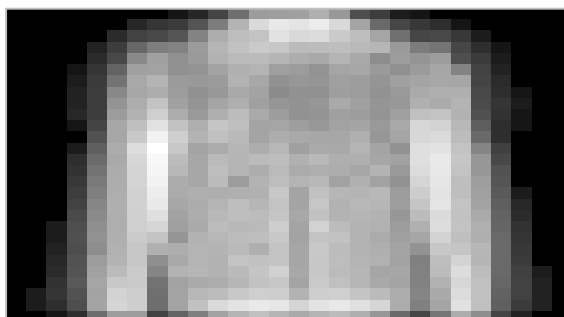
MiSo



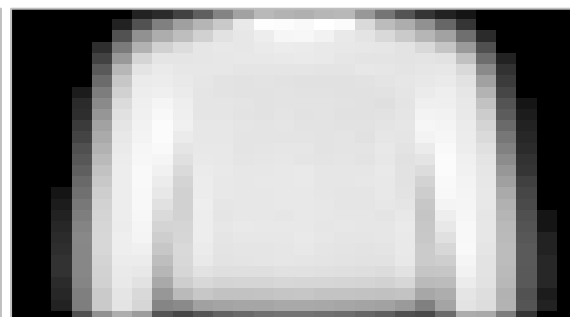
Shirt: observed



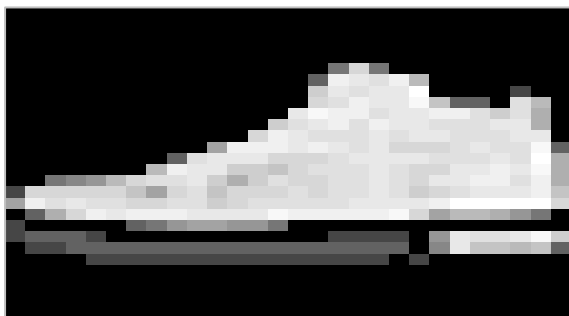
Vanilla NMF



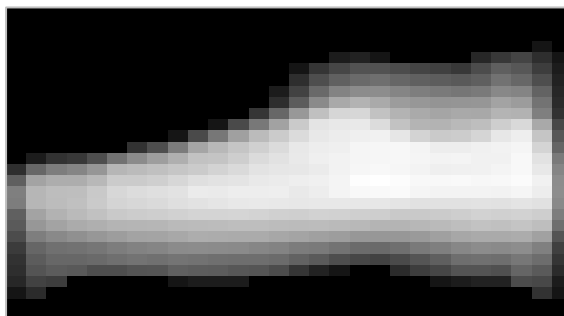
MiSo



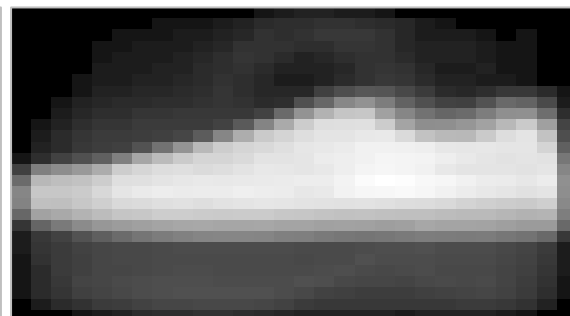
Sneaker: observed



Vanilla NMF



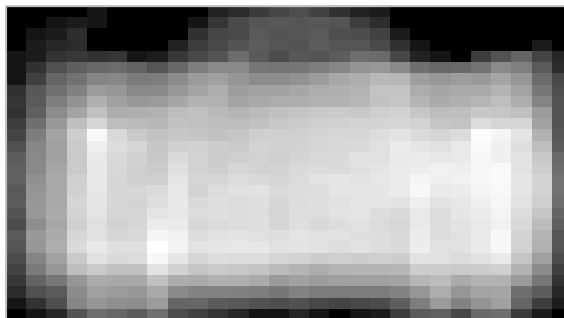
MiSo



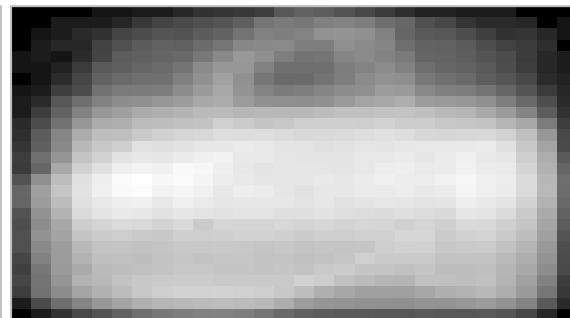
Bag: observed



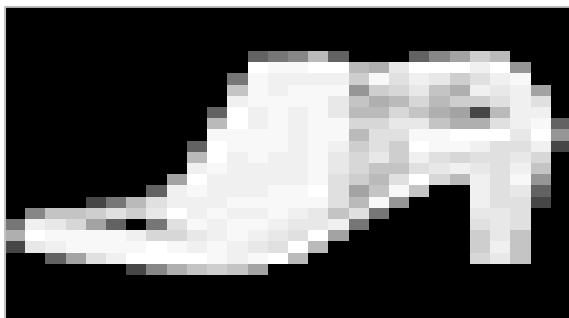
Vanilla NMF



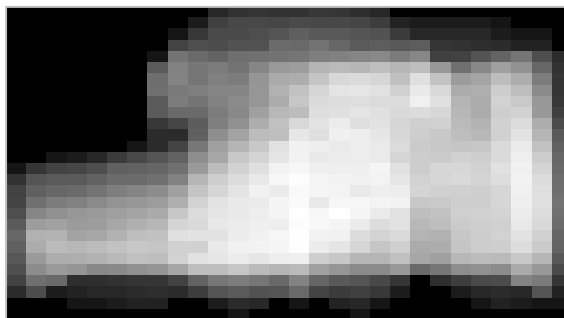
MiSo



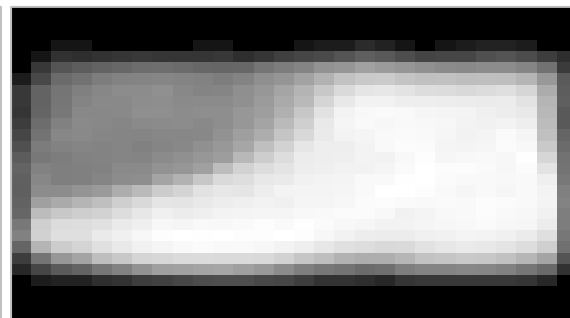
Ankle boot: observed



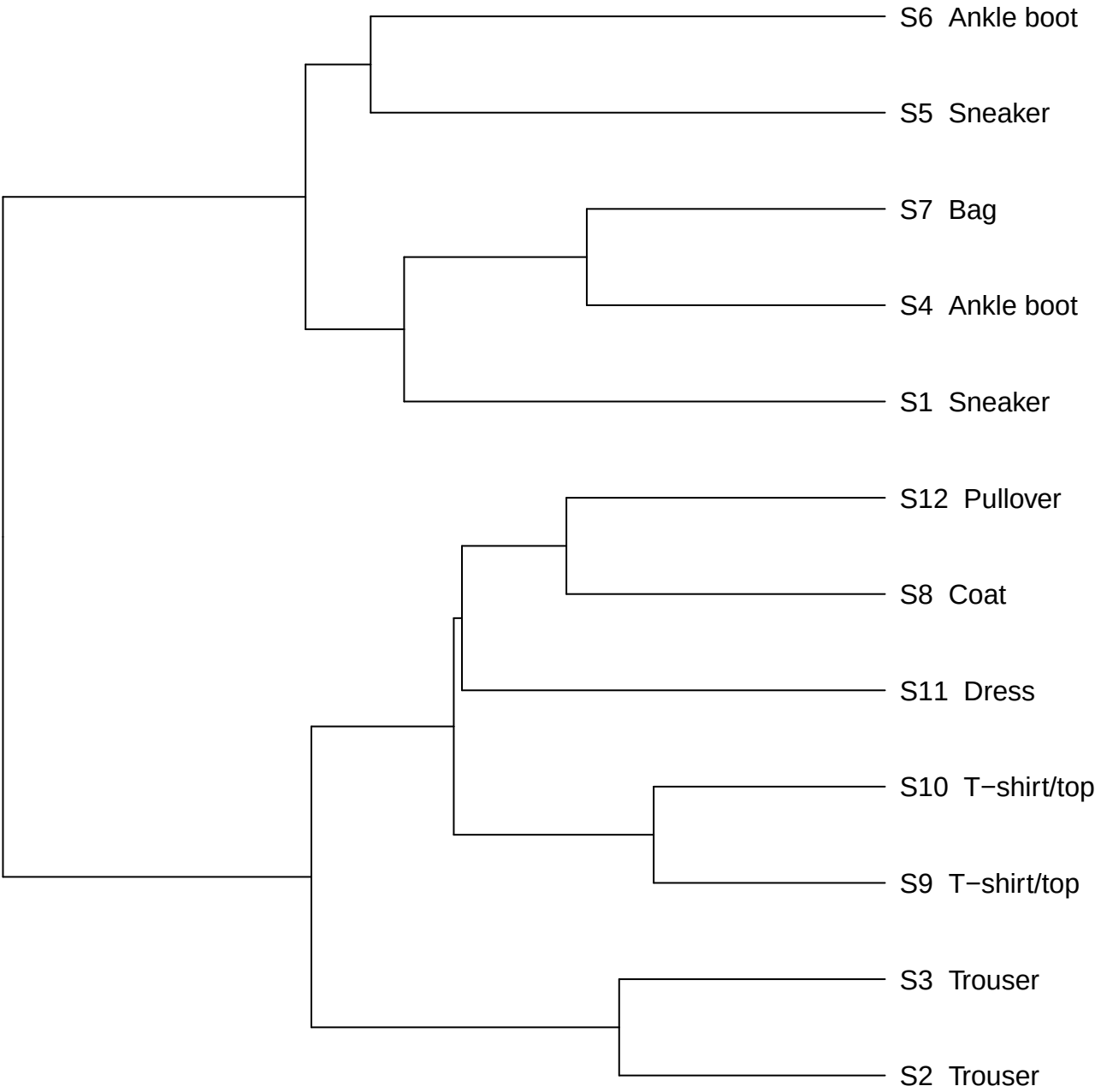
Vanilla NMF



MiSo



Submanifold tree from Sinkhorn/Wasserstein distance



0.4

0.3

0.2

0.1

0.0

Factor graph: edges show co-use within fitted submanifolds

